ผลลัพธ์แสดง Gateway of last resort เป็น 0.0.0.0 to network 0.0.0.0 ยืนยันว่า IPv4 static default route ทำงานถูกต้อง โดยเส้นทางหลักผ่าน Serial0/0/0 ไปยัง ISP1 ส่วน floating route ผ่าน Serial0/0/1 ยังไม่ปรากฏเพราะมี AD สูงกว่า (5) นอกจากนี้ยังแสดง host route ไปยัง Customer Server (198.0.0.10/32) และIPv6 static default route (::/0) ผ่าน ISP1 เป็นเส้นทางหลัก พร้อม host route ไปยัง Customer Server (2001:db8:f:f::10/128) ส่วน floating route ผ่าน ISP2 (AD=5) ยังไม่แสดง เนื่องจากเส้นทางหลักยังใช้งานได้อยู่

Edge_Router

CLI

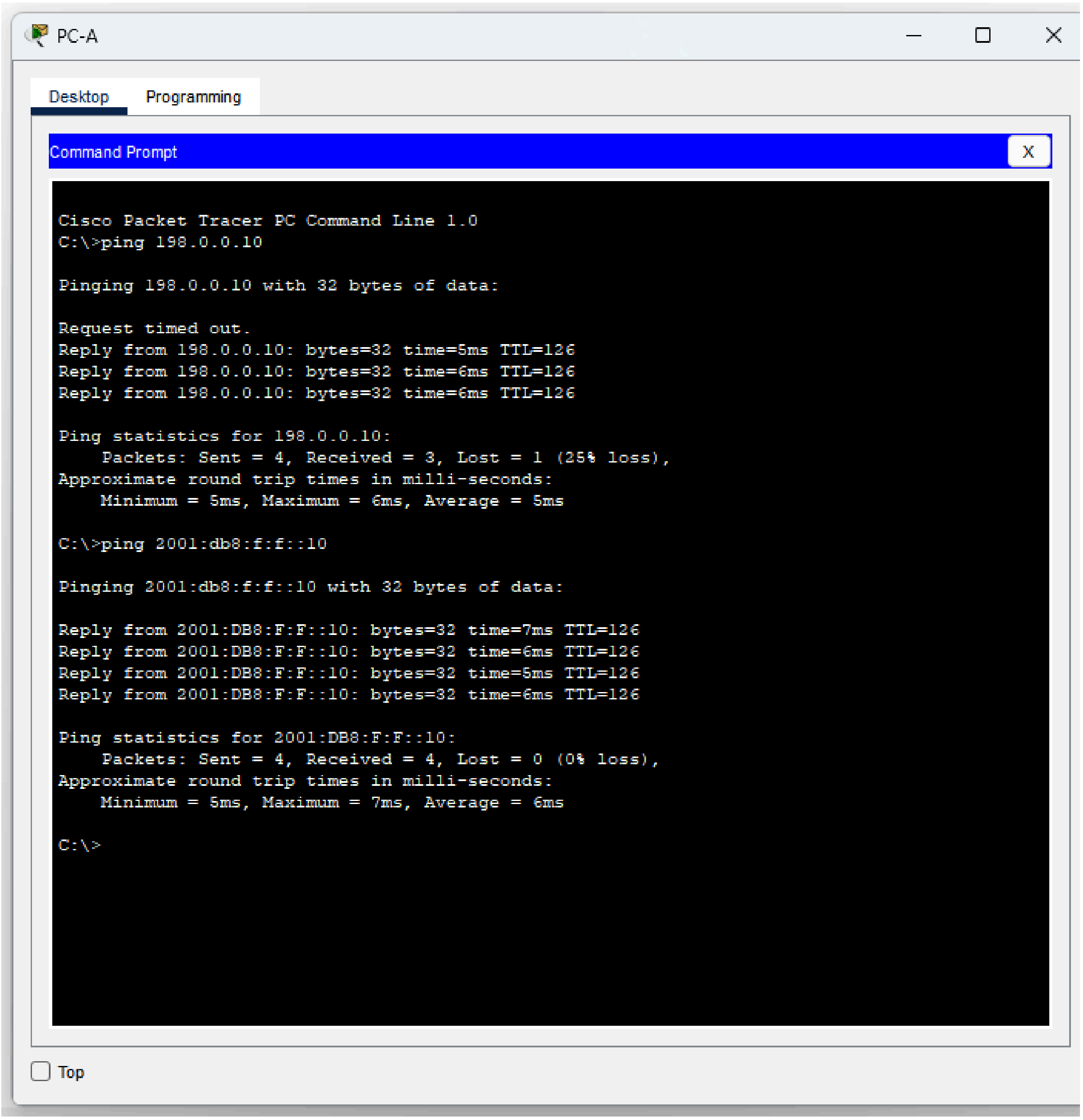
```
Edge_Router#
Edge_Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.10.10.0/30 is directly connected, Serial0/0/0
L       10.10.10.2/32 is directly connected, Serial0/0/0
C       10.10.10.4/30 is directly connected, Serial0/0/1
L       10.10.10.6/32 is directly connected, Serial0/0/1
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.16/28 is directly connected, GigabitEthernet0/0
L       192.168.10.17/32 is directly connected, GigabitEthernet0/0
     192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.11.32/27 is directly connected, GigabitEthernet0/1
L       192.168.11.33/32 is directly connected, GigabitEthernet0/1
     198.0.0.0/32 is subnetted, 1 subnets
S       198.0.0.10/32 is directly connected, Serial0/0/0
S*   0.0.0.0/0 is directly connected, Serial0/0/0

Edge_Router#show ipv6 route
IPv6 Routing Table - 11 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
S   ::/0 [1/0]
     via 2001:DB8:A:1::1
C   2001:DB8:1:10::/64 [0/0]
     via GigabitEthernet0/0, directly connected
L   2001:DB8:1:10::1/128 [0/0]
     via GigabitEthernet0/0, receive
C   2001:DB8:1:11::/64 [0/0]
     via GigabitEthernet0/1, directly connected
L   2001:DB8:1:11::1/128 [0/0]
     via GigabitEthernet0/1, receive
C   2001:DB8:A:1::/64 [0/0]
     via Serial0/0/0, directly connected
L   2001:DB8:A:1::2/128 [0/0]
     via Serial0/0/0, receive
C   2001:DB8:A:2::/64 [0/0]
     via Serial0/0/1, directly connected
L   2001:DB8:A:2::2/128 [0/0]
     via Serial0/0/1, receive
S   2001:DB8:F:F::10/128 [1/0]
     via 2001:DB8:A:1::1
L   FF00::/8 [0/0]
     via Null0, receive
Edge_Router#
```

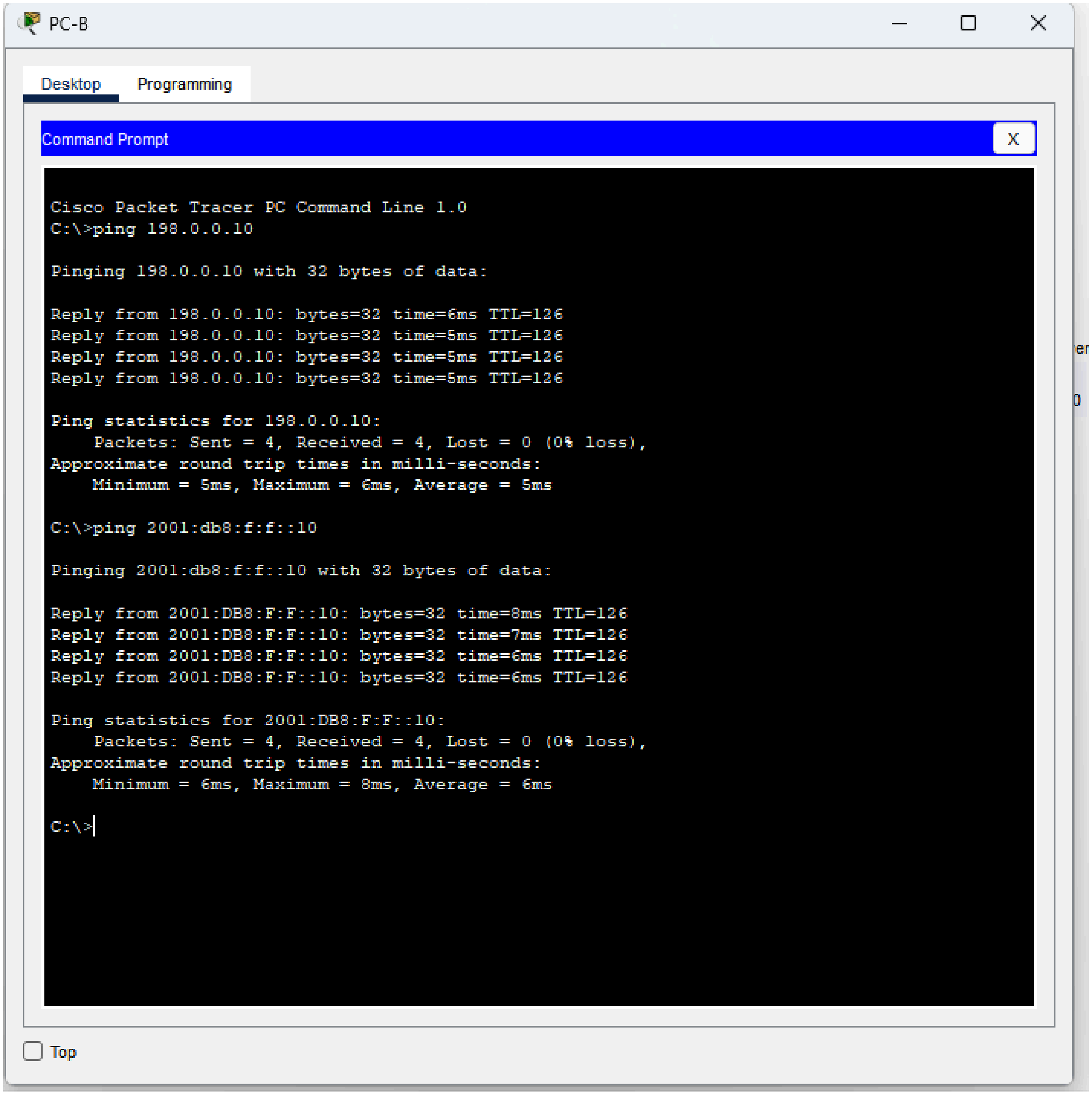
ผลลัพธ์แสดง static route ไปยัง LAN 1 (2001:db8:1:10::/64) และ LAN 2 (2001:db8:1:11::/64) ด้วย AD=1 ผ่าน Edge_Router (2001:db8:a:1::2) เป็นเส้นทางหลัก ส่วน floating route ผ่าน ISP2 (AD=5) ยังไม่ปรากฏเนื่องจากเส้นทางหลักยังทำงานปกติ

```
ISP1#
ISP1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.10.10.0/30 is directly connected, Serial0/0/0
L       10.10.10.1/32 is directly connected, Serial0/0/0
     192.168.10.0/28 is subnetted, 1 subnets
S       192.168.10.16/28 [1/0] via 10.10.10.2
     192.168.11.0/27 is subnetted, 1 subnets
S       192.168.11.32/27 [1/0] via 10.10.10.2
     198.0.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       198.0.0.0/24 is directly connected, GigabitEthernet0/0
L       198.0.0.1/32 is directly connected, GigabitEthernet0/0

ISP1#show ipv6 route
IPv6 Routing Table - 7 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
S   2001:DB8:1:10::/64 [1/0]
     via 2001:DB8:A:1::2
S   2001:DB8:1:11::/64 [1/0]
     via 2001:DB8:A:1::2
C   2001:DB8:A:1::/64 [0/0]
     via Serial0/0/0, directly connected
L   2001:DB8:A:1::1/128 [0/0]
     via Serial0/0/0, receive
C   2001:DB8:F:F::/64 [0/0]
     via GigabitEthernet0/0, directly connected
L   2001:DB8:F:F::1/128 [0/0]
     via GigabitEthernet0/0, receive
L   FF00::/8 [0/0]
     via Null0, receive
ISP1#
```

ภาพ: Ping จาก PC-A และ PC-B ไปยัง Customer Server (เส้นทางหลัก)

ผลการทดสอบ ping จาก PC-A และ PC-B ไปยัง Customer Server ทั้ง IPv4 (198.0.0.10) และ IPv6 (2001:db8:f:f::10) สำเร็จทั้งหมด (Success rate 100%) ยืนยันว่า static route และ default route ที่ตั้งค่าไว้ทำงานถูกต้องผ่านเส้นทางหลัก




PC-B

Desktop Programming

Command Prompt X

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 198.0.0.10

Pinging 198.0.0.10 with 32 bytes of data:

Reply from 198.0.0.10: bytes=32 time=6ms TTL=126
Reply from 198.0.0.10: bytes=32 time=5ms TTL=126
Reply from 198.0.0.10: bytes=32 time=5ms TTL=126
Reply from 198.0.0.10: bytes=32 time=5ms TTL=126

Ping statistics for 198.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 6ms, Average = 5ms

C:\>ping 2001:db8:f:f::10

Pinging 2001:db8:f:f::10 with 32 bytes of data:

Reply from 2001:DB8:F:F::10: bytes=32 time=8ms TTL=126
Reply from 2001:DB8:F:F::10: bytes=32 time=7ms TTL=126
Reply from 2001:DB8:F:F::10: bytes=32 time=6ms TTL=126
Reply from 2001:DB8:F:F::10: bytes=32 time=6ms TTL=126

Ping statistics for 2001:DB8:F:F::10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 6ms, Maximum = 8ms, Average = 6ms

C:\>
```

Top

หลังจาก shutdown interface Serial0/0/0 บน Edge_Router เพื่อจำลองเส้นทางหลักไปยัง ISP1 ขาดการเชื่อมต่อ ผลการทดสอบ ping จาก PC-A และ PC-B ไปยัง Customer Server ยังคงสำเร็จ แสดงว่า floating static route ผ่าน ISP2 เริ่มทำงานทดแทนโดยอัตโนมัติ ยืนยันความสามารถด้าน redundancy ของระบบตามวัตถุประสงค์ของแลป

```
Edge_Router#enable
Edge_Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Edge_Router(config)#interface Serial0/0/0
Edge_Router(config-if)#shutdown

Edge_Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to down
end
Edge_Router#
%SYS-5-CONFIG_I: Configured from console by console
```

```
C:\>ping 198.0.0.10

Pinging 198.0.0.10 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 198.0.0.10: bytes=32 time=6ms TTL=125
Reply from 198.0.0.10: bytes=32 time=6ms TTL=125

Ping statistics for 198.0.0.10:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 6ms, Maximum = 6ms, Average = 6ms

C:\>ping 2001:db8:f:f::10

Pinging 2001:db8:f:f::10 with 32 bytes of data:

Reply from 2001:DB8:F:F::10: bytes=32 time=5ms TTL=125
Reply from 2001:DB8:F:F::10: bytes=32 time=4ms TTL=125
Reply from 2001:DB8:F:F::10: bytes=32 time=5ms TTL=125
Reply from 2001:DB8:F:F::10: bytes=32 time=5ms TTL=125

Ping statistics for 2001:DB8:F:F::10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 5ms, Average = 4ms

C:\>
```